Dataset 2: Brain Network

Description

The data is a matrix consisting of resting state fMRI data obtained from over 1,000 participants by Biswal et al. (2010). It is a measure of brain activity while the subject is at rest. The matrix is constructed by parceling the brain into different regions based on a brain atlas, and then obtaining the Pearson correlation of activation across the regions. This is done for each participant, and then averaged into a single matrix which is provided to you.

Datasets

- `AveragedMatrix.txt`: contains the averaged connectivity matrix from over 1,000 participants
- `region_names_abbrev_file.txt`: names of brain regions (nodes), abbreviated
- `region_names_full_file.txt`: names of brain regions (nodes), full names

Note: The full names contain duplicates, such as "Left Frontal Pole", but in the short names you will see that there are 6 different regions associated with this general brain area (LFP1, LFP2... LFP6).

Helper Script: `dataset2.R`

Contains code for additional clean up of `AveragedMatrix.txt`, adding of node/brain region labels, and conversion to a suitable format for further network analysis.

References

Biswal, B. B., Mennes, M., Zuo, X.-N., Gohel, S., Kelly, C., Smith, S. M., Beckmann, C. F., Adelstein, J. S., Buckner, R. L., Colcombe, S., Dogonowski, A.-M., Ernst, M., Fair, D., Hampson, M., Hoptman, M. J., Hyde, J. S., Kiviniemi, V. J., Kötter, R., Li, S.-J., ... Milham, M. P. (2010). Toward discovery science of human brain function. *Proceedings of the National Academy of Sciences, 107*(10), 4734–4739. https://doi.org/10.1073/pnas.0911855107

The raw data was obtained from this website: http://umcd.humanconnectome project.org/umcd/default/browse_studies